the quotient of df and dx.Well, in some cases yes. For instance, if $f(x) = mx + c, m, c$ are constants, then, $y = f(x)$.

$$\Delta y = f(x+\Delta x) - f(x) = m\Delta x = f'(x)\Delta x \quad \text{for all } x \in \mathbb{R} \text{ and } \Delta x$$

and hence equality in both (2), and (3). In this case changes in x and $y(= f)$ are taking place along straight lines, in which case we have,

$$\frac{\text{change in } f}{\text{change in } x} = \frac{\Delta y}{\Delta x} = f'(x) = \frac{df}{dx} = \frac{dy}{dx}.$$

Thus in this case the derivative $\frac{df}{dx}$ is truly a quotient of df and dx, if we take $df = \Delta f = dy$ and $dx = \Delta x$. This leads us to define the differential of f as follows:

Definition 8.4

Let $f : (a,b) \to \mathbb{R}$ be a differentiable function, for $x \in (a,b)$ and Δx the increment given to x, we define the differential of f by

$$df = f'(x)\Delta x. \qquad \text{... (8)}$$

First we note that if $f(x) = x$, then by (8) we get $dx = f'(x)\Delta x = 1\Delta x$ which means that the differential $dx = \Delta x$, which is the change in x-axis. So the differential given by (8) is same as $df = f'(x)dx$.

Next we explore the differential for an arbitrary differentiable function $y = f(x)$. Then $\Delta f = f(x+dx) - f(x)$ gives the change in output along the graph of $y = f(x)$ and $f'(x)$ gives the slope of the tangent line at $(x, f(x))$. Let dy or df denote the increment in f along the tangent line. Then by the above observation, we have $dy = f'(x)dx$.

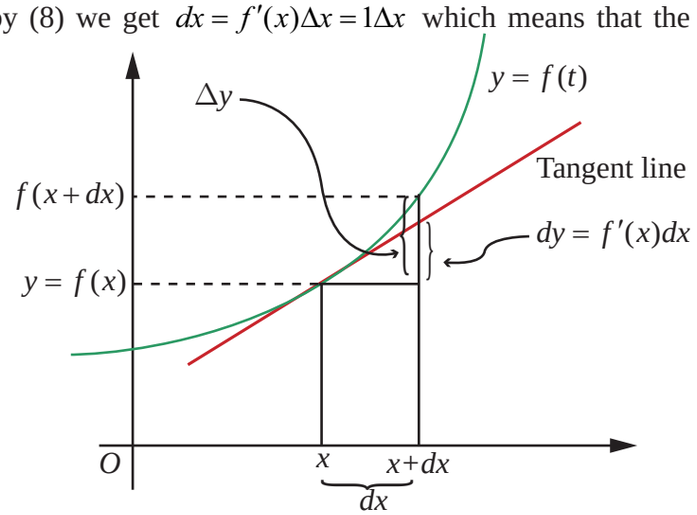


Fig. 8.4

Linear Approximation and Differential

From the figure it is clear that $\Delta f \approx dy = df = f'(x)dx$ and hence $f'(x)$ can be viewed approximately as the quotient of Δf and Δx. So we may interpret $\frac{df}{dx}$ as the quotient of df and dx.

Remark

We know that derivative of a function is again a function. On the other hand, differential df of a function f is not only a function of the independent variable but also depends on the change in the input namely $dx = \Delta x$. So df is a function of two changing quantities namely x and dx. Observe that $\Delta f \approx df$, which can be observed from the Fig. 8.4.

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